Play and Explore

|Programming with Arduino|



|Programming with Arduino|

Example 4

TRY IT OUT!

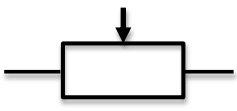
Potentiometer

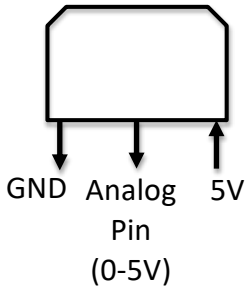


Potentiometer

A potentiometer is a simple knob that provides a variable resistance, which we can read into the Arduino board as an analog value.

Electrical symbol:

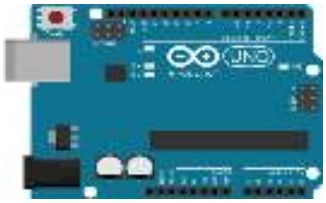






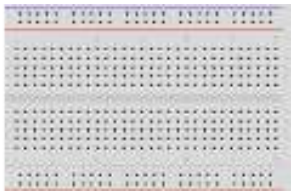
Example 4

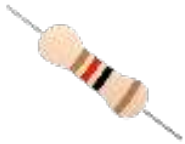
Equipment



- Arduino
- Breadboard
- Potentiometer
- LED
- Resistor, $R = 330\Omega$
- Wires (Jumper Wires)



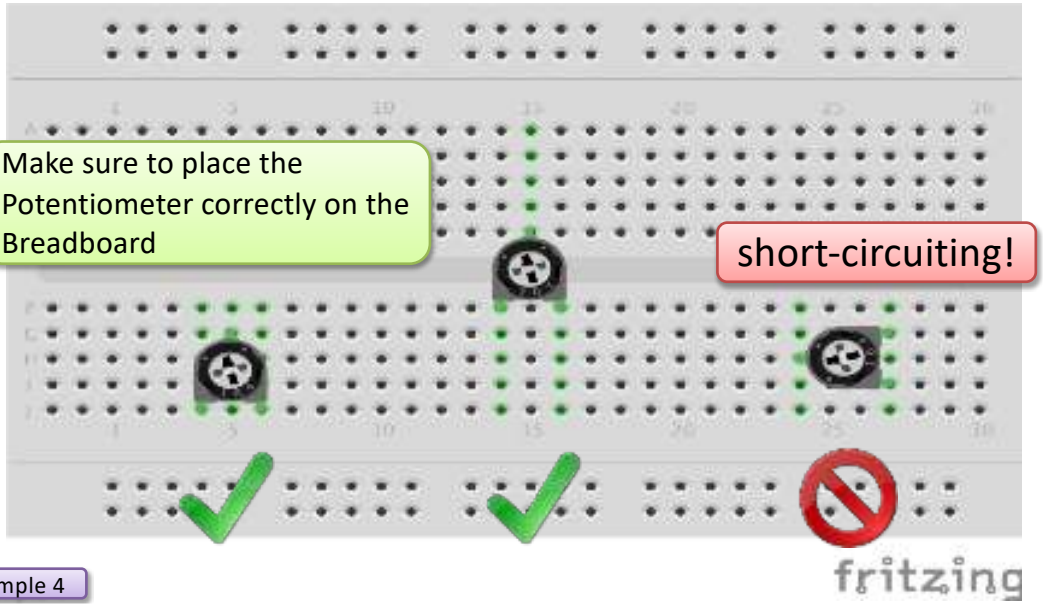




Example 4

Breadboard





Example 4

Dimmer



In this example we will make a simple dimmer using a potentiometer that control the intensity of the light.

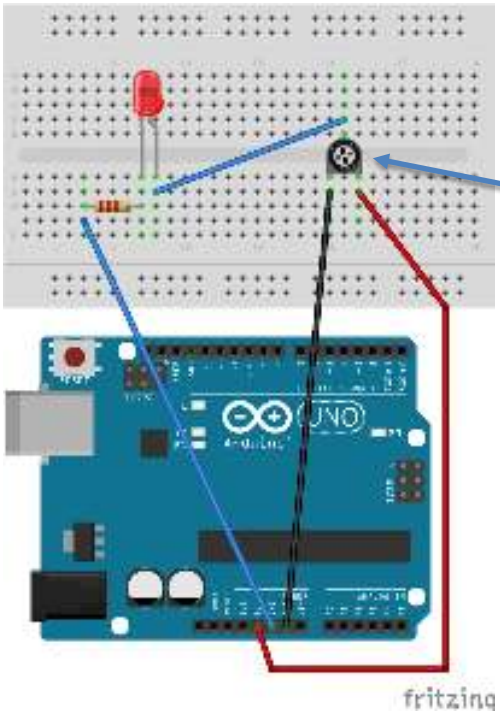
This is a typical example where a potentiometer is used. Everybody have a dimmer at home.

When the voltage in the circuit increases, the intensity of the LED will increase.

Example 4

Note! No Arduino Program is needed in this example

Wiring



TRY IT OUT!

Adjust the the dimmer (potentiometer) in order to increase or decrease the intensity of the light (LED)

Note! No Arduino Program is needed in this example

Example 4

Play and Explore

|Programming with Arduino|



|Programming with Arduino|

Example 5

TRY IT OUT!

Temperature



Introduction

In this example we will use a small temperature sensor to read the temperature in the room.

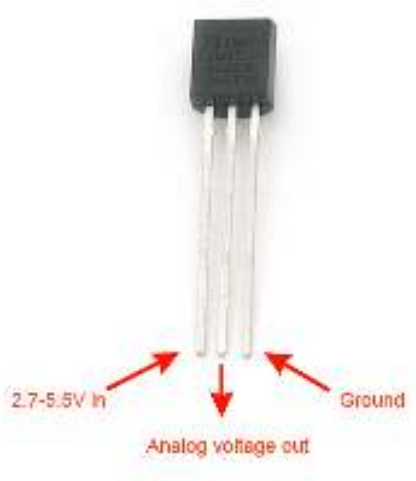
In this example we will use one of the "Analog In" ports on the Arduino board



Example 5

TMP36 Temperature Sensor





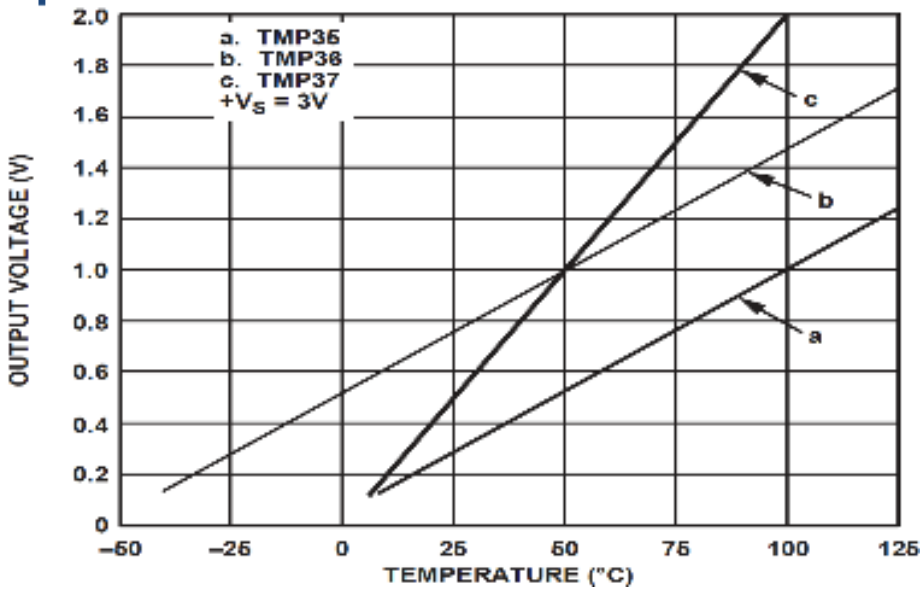
A Temperature sensor like TM36 use a solid-state technique to determine the temperature.

They use the fact as temperature increases, the voltage across a diode increases at a known rate.

Example 5

https://learn.adafruit.com/tmp36-temperature-sensor

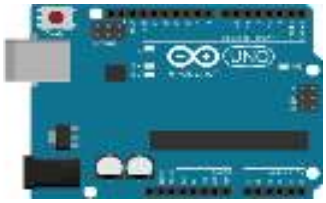
Temperature Sensor Datasheet



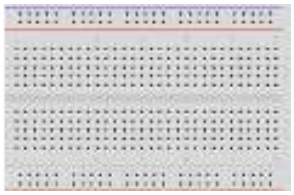
Example 5

[http://no.rs-online.com/webdocs/14cd/0900766b814cd0a1.pdf]

Necessary Equipment







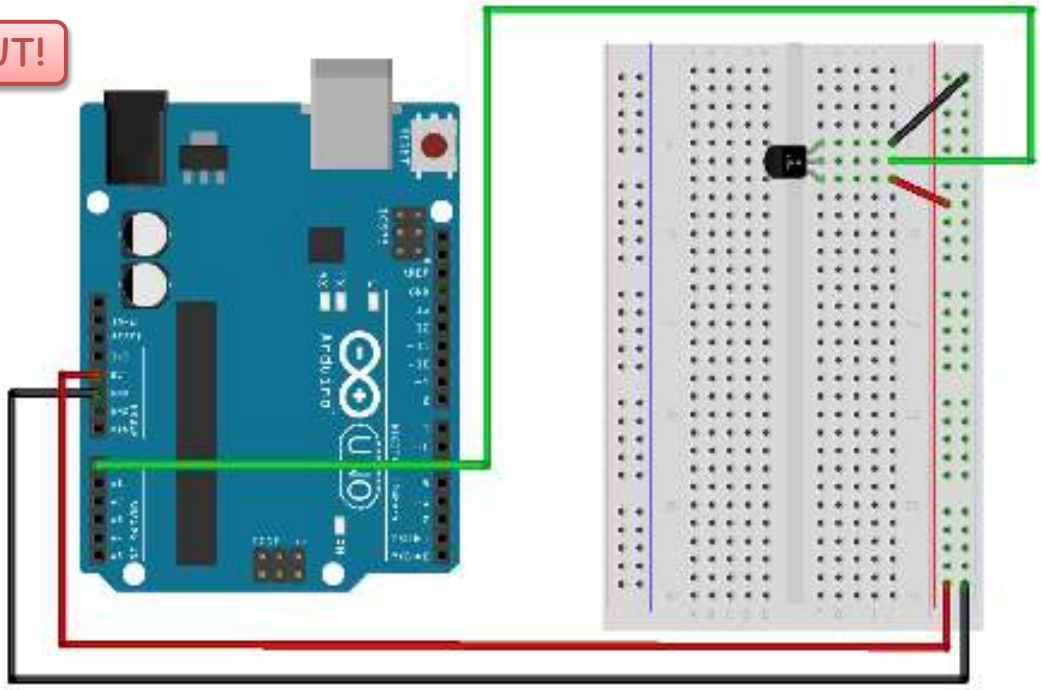
- Arduino
- Breadboard
- TMP36
- Wires (Jumper Wires)



Example 5

TRY IT OUT!

Wiring



Example 5

analogRead

analogRead reads the value from a specific analog pin.

The Arduino UNO board has 6 analog pins (channels), and uses a 10-bit analog to digital converter.

Syntax:

value = analogRead(analogPin);

value will then be between 0 and 1023

Example:

```
int sensorPin = 0;
int sensorValue;

void setup()
{

}

void loop()
{
  sensorValue = analogRead(sensorPin);
}
```

Example 5

https://www.arduino.cc/en/Reference/AnalogRead

Temperature coversion

We want to present the value from the sensor in degrees Celsius:

1. analogRead() gives a value between 0 and 1023
2. Then we convert this value to 0-5V
3. Finally, we convert to degrees Celsius using information from the Datasheet presented on the previous page

Example 5

Arduino Program

Example 5

```
const int temperaturePin = 0;

float adcValue;
float voltage;
float degreesC;

void setup()
{
  Serial.begin(9600);
}

void loop()
{

  adcValue = analogRead(temperaturePin);

  voltage = (adcValue*5)/1023;

  degreesC = 100*voltage - 50;

  Serial.print("ADC Value: ");
  Serial.print(adcValue);

  Serial.print("  voltage: ");
  Serial.print(voltage);

  Serial.print("  deg C: ");
  Serial.println(degreesC);

  delay(1000);
}
```

TRY IT OUT!

Convert from ADC-value (0-1023) to Voltage (0-5V)

Convert from Volate to degrees Celsius

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|Programming with Arduino|

Example 6

TRY IT OUT!

Light Sensor



Introduction

In this example we will use a light sensor to measure the light intensity of the room.

If it's dark, we will turn on the light (LED)
If it's bright, we'll turn off the light (LED)



Example 6

Light Sensor


or

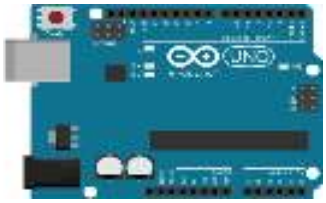

Light sensor, Photocell (Photo resistor), LDR (light-dependent resistor)

A light sensor / photocell is a sensor used to detect light.

The resistance decreases with increasing light intensity (stronger light).

Example 6

Necessary Equipment



Light Sensor

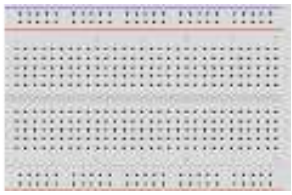


or





LED

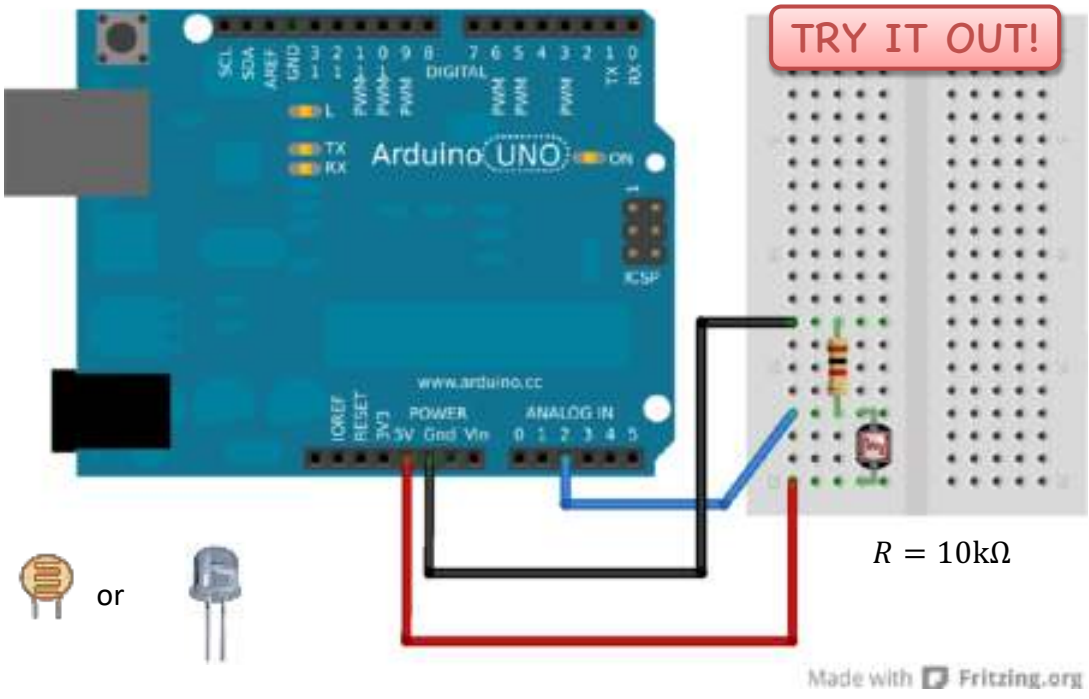


- Arduino
- Breadboard
- Light Sensor
- LED
- Resistors, $R = 330\Omega$, $R = 10\ k\Omega$
- Wires (Jumper Wires)



Example 6

Wiring



or

Example 6

Arduino Program

Example 6

```
int photocellPin = 2;
int photocellReading;

void setup(void)
{
    Serial.begin(9600);
}

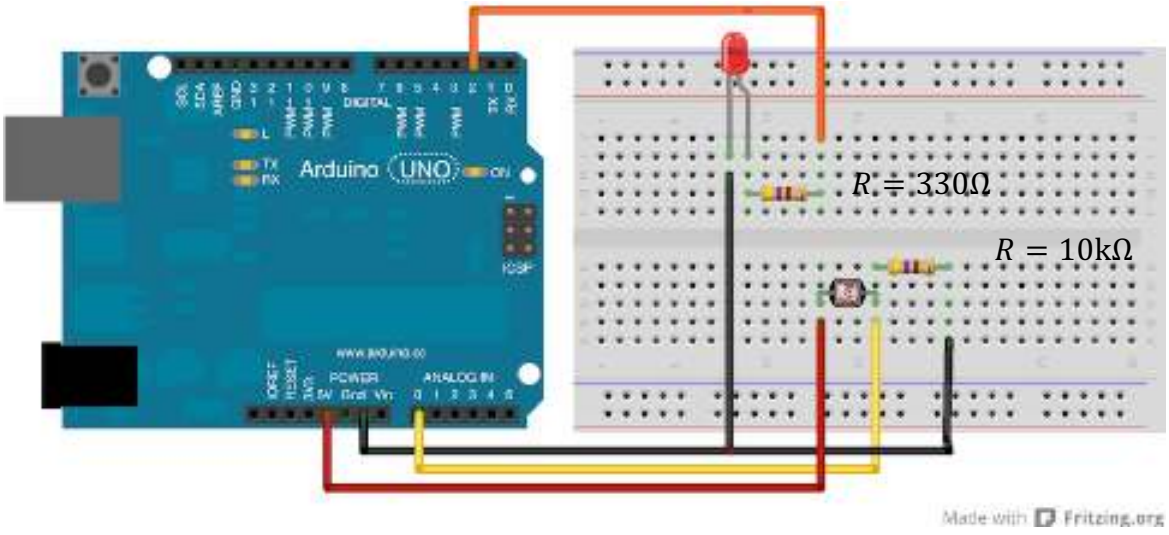
void loop(void)
{
    photocellReading = analogRead(photocellPin);

    Serial.print("Analog reading = ");
    Serial.println(photocellReading);

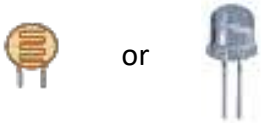
    delay(1000);
}
```

TRY IT OUT!

Wiring



or



Example 6

Arduino Program

Example 6

```
int photocellPin = 0;
int ledPin = 2;
int photocellReading;
const float limit = 100;

void setup(void)
{
      Serial.begin(9600);
      pinMode(ledPin, OUTPUT);
}

void loop(void)
{
      photocellReading = analogRead(photocellPin);
      Serial.print("Analog reading = ");
      Serial.println(photocellReading);

      if (photocellReading < limit)
      {
            digitalWrite(ledPin, HIGH);
      }
      else
      {
            digitalWrite(ledPin, LOW);
      }

      delay(1000);

}
```

TRY IT OUT!

Play and Explore

|Programming with Arduino|



|Programming with Arduino|

Example 7

TRY IT OUT!

Thermistor



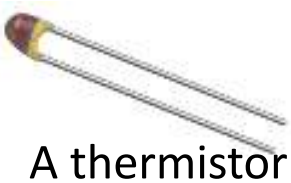
Introduction

In this example we will use a small thermistor to read the temperature in the room.

In this example we will use one of the "Analogue Inn" ports at Arduino.



Example 7



Thermistor

A thermistor is an electronic component that changes resistance to temperature - so-called Resistance Temperature Detectors (RTD). It is often used as a temperature sensor.

Our Thermistor is a so-called NTC (Negative Temperature Coefficient). In a NTC Thermistor, resistance decreases as the temperature rises.

There is an non-linear relationship between resistance and excitement. To find the temperature we can use the following equation (Steinhart-Hart equation):

$$\frac{1}{T} = A + B\ln(R) + C(\ln(R))^3$$

where A, B, C are constants given below [Wikipedia]

$A = 0.001129148, B = 0.000234125\ and\ C = 8.76741E-08$

Example 7



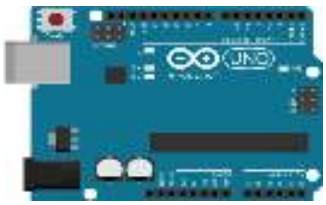
NTC Thermistor - Datasheet

Technical data	
Resistance @ 25°C	10 kΩ
Temperature range	-40...+125 °C
Power max.	500 mW
Pitch	2.54 mm
Resistance tolerance	±5 %
$W_{25/100}$ value	3977 K
B value tolerance	±0.75 %
Thermal time constant	15 s

Datasheet: https://www.elfadistrelec.no/no/ntc-motstand-kablet-10-kohm-vishay-ntcle100e3103jb0/p/16026041?q=160-26-041&page=1&origPos=1&origPageSize=50&simi=98.0

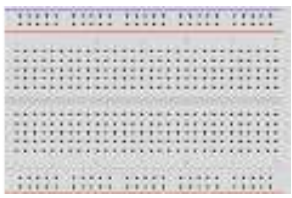
Example 7

Equipment







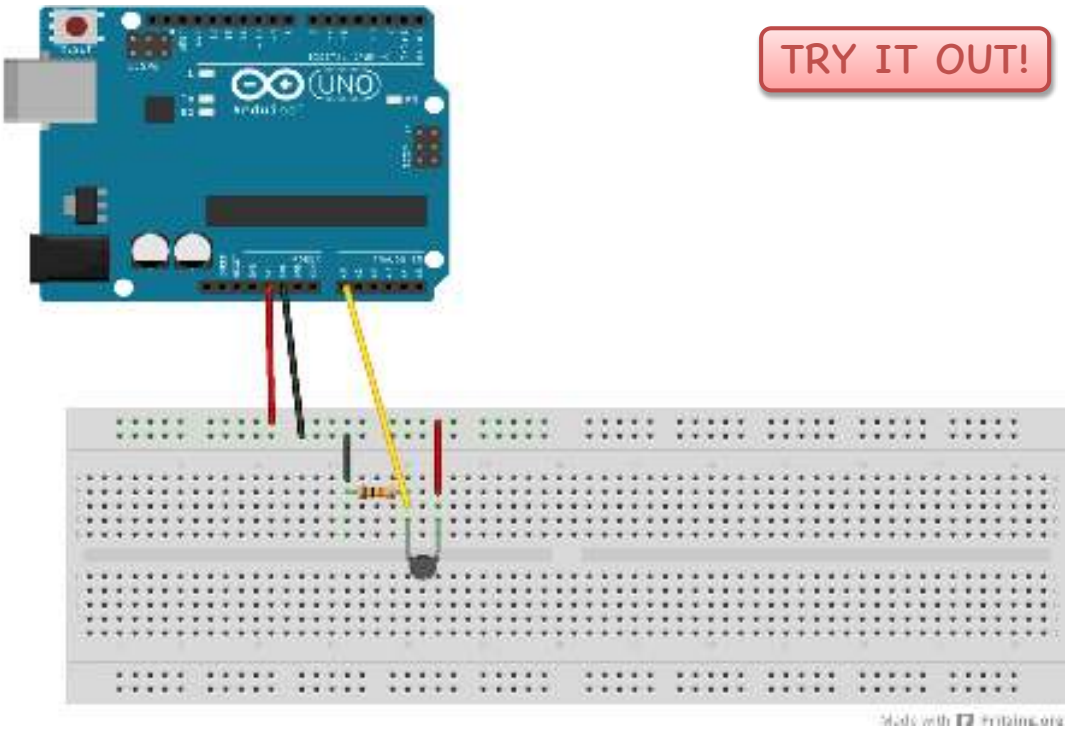




- Arduino
- Breadboard
- Thermistor
- LED
- Resistor 10 kΩ
- Wires (Jumper Wires)

Example 7

Wiring

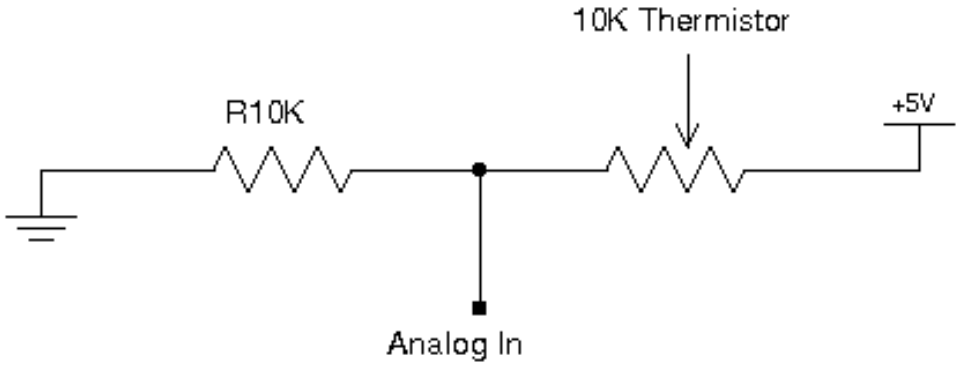


TRY IT OUT!

Example 7

The wiring is called a Voltage divider:

Wiring



[https://en.wikipedia.org/wiki/Voltage_divider]

Example 7

Arduino Program

TRY IT OUT!

```
const int temperaturePin = 0;

void setup()
{
  Serial.begin(9600);
}

void loop()
{
  int temperature = getTemp();
  Serial.print("Temperature Value: ");
  Serial.print(temperature);
  Serial.println("*C");
  delay(1000);
}

double getTemp()
{
  // Inputs ADC Value from Thermistor and outputs Temperature in Celsius

  int RawADC = analogRead(temperaturePin);
  long Resistance;
  double Temp;

  // Assuming a 10k Thermistor.  Calculation is actually: Resistance = (1024/ADC)
  Resistance=((10240000/RawADC) - 10000);

  // Utilizes the Steinhart-Hart Thermistor Equation:
  // Temperature in Kelvin = 1 / {A + B[ln(R)] + C[ln(R)]^3}
  // where A = 0.001129148, B = 0.000234125 and C = 8.76741E-08

  Temp = log(Resistance);
  Temp = 1 / (0.001129148 + (0.000234125 * Temp) + (0.0000000876741 * Temp * Temp * Temp));
  Temp = Temp - 273.15;  // Convert Kelvin to Celsius
  return Temp;  // Return the Temperature
}
```

Example 7

Additional Exercises

TRY IT OUT!

Try out the following:

- Use both the TMP36 Temperature Sensor we used earlier and the Thermistor.
- Compare the values from the Thermistor with the TMP36 Temperature Sensor we used earlier. Display both values in the Serial Monitor.
- Draw circuit and wiring using the Fritzing software

Example 7

Play and Explore

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|Slutt på eksemplet|

Play and Explore

|Programming with Arduino|

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